

Bc. Nina Duchyňová

MASTER'S STUDENT IN ASTROPHYSICS

CONFERENCES

Meeting of members of Czech Astronomical Society

Brno Observatory and Planetarium
15.-16. April 2023

Conference on variable star research

Ostrava Planetarium
25.-26. November 2022

VOLUNTEERING

Student Volunteer 12.-17. May 2024

STARMUS Festival, Bratislava, Slovakia

- Volunteered at the Starmus Festival in Bratislava, representing Masaryk University by engaging the public with astronomy through telescope observations and science discussions.

Blood donor 2022 - Present

Red Cross Blood Donor

SKILLS

Programming Languages: Python - NumPy, Pandas, Matplotlib, Astropy, Astroquery

Tools: LaTeX, Jupyter Notebook, Visual Studio Code, Ubuntu, SAOImage DS9, IRAF, GALFIT, QtiPlot, TOPCAT, DCR, TinyTim, MS Excel, MS Word, Canva

Soft skills: Coordinating small groups before and during outreach events, open communication and feedback, independent and collaborative work approach

LANGUAGE

- Slovak native
- Czech native/bilingual
- English full professional
- German elementary

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Personal website

Bratislava, Slovakia

EDUCATION

Astrophysics Masaryk University Master's program 2025 - Present

- First year master's student with diploma thesis focused on Tidal Disruption Events as an explanation for Little Red Dots, under the supervision of Isobelle Lilian Mary Garland, Ph.D. and RNDr. Michal Zajaček, Dr. rer. nat.

Astrophysics Masaryk University Bachelor's program 2022 - 2025

- Thesis: Why do not all radio galaxies form extended structures?
We investigate the differences between compact and extended radio sources by analysing their physical and host galaxy properties, based on the ROGUE catalogue. We classified galaxies with diagnostic diagrams (DLM, BPT) to distinguish AGN and star-forming systems, and compare radio and far-infrared luminosity to assess radio excess. All done by self made Python code.

EXPERIENCE

CSO Host Galaxy Research Project Autumn 2025 Faculty of science, Masaryk University Brno, Czech Republic

- Investigating luminosity profiles of galaxies hosting compact symmetric radio sources using HST archival imaging and GALFIT modelling to probe potential links to tidal disruption events. Built the workflow by independently adopting IRAF, GALFIT, TinyTim, and DCR for data reduction and fitting.

Research Intern July 2025 - September 2025 Nicolaus Copernicus Astronomical Center, Warsaw, Poland Polish Academy of Sciences

- Project: Do primordial black holes exist?
Supervised by Prof. Marek Abramowicz and assistance from Dr. Maciek Wielgus in Warsaw, this summer project focused on the possible existence of primordial black holes. My work involved literature review, improving and testing a Python-based population synthesis code, and performing basic calculations and programming related to neutron-star collisions.

Outreach & Events Organisation September 2023 - Present Faculty of Science, Masaryk University Brno, Czech Republic

- Involved in faculty's science outreach, helping organize and take part in events such as Researchers' Night, the Science and Technology Festival, Open Day, Gaudeamus and more. I also partially assist in producing our department's podcast "Rozhovory o vesmíru", contributing to video editing and background research. I also gave and helped organize a lecture for children on Cosmology through the Department's Astronomy Club and participated in MJUNI, where I helped with a lecture on Dangers from the Cosmos.

Customer service assistant September 2019 - August 2021 Panta Rhei s.r.o. Bratislava, Slovakia

- Provided customer service, resolved issues, answered inquiries, and provided product information. Engaged in conversation with customers to better uncover individual needs and maintained a well-organised and tidy store environment.